WaveSplat: Grid-Free Spectral Gaussian Wave Field Estimation from Sparse LiDAR

Saeyong Park
Avikus (HD Hyundai)
saeyong.park@avikus.ai

***Abstract*—We present WaveSplat, a grid-free method for real-time ocean wave field estimation from sparse shipboard LiDAR observations. Unlike grid-based approaches such as the Ensemble Transform Kalman Filter (ETKF), WaveSplat represents the wave field as a weighted sum of spectral Gaussian components—plane waves parameterized by wavenumber, amplitude, and phase—with angular frequency constrained by the linear dispersion relation. This grid-free representation naturally avoids interpolation artifacts, scales to arbitrary spatial resolution, and enables an adaptive number of wave components matched to scene complexity. We propose a two-phase estimation pipeline: (1) initialization via dispersion-aware beamforming with coarse-to-fine refinement, and (2) per-frame tracking with physics-based phase prediction and selective gradient updates. Fresnel-zone confidence weighting accounts for range- and angle-dependent LiDAR measurement noise. On 12 representative episodes spanning Beaufort 2–5 sea states (1–30 wave components), WaveSplat achieves an average nRMSE of 0.211 compared to 0.436 for ETKF-grid (51.5% improvement), while running 2.4× faster (13.4 vs. 31.7 ms/frame). With adaptive N, the model automatically selects N=1 for simple swells and N=21 for complex multi-component seas.**

***Index Terms*—Wave field estimation, LiDAR, spectral methods, Gaussian splatting, dispersion relation, Kalman filter**

I. Introduction

Accurate estimation of the ocean wave field around a vessel is critical for autonomous ship operations, including motion prediction, route optimization, and structural load monitoring [1]. Shipboard scanning LiDAR has emerged as a practical sensor for this task, providing direct elevation measurements over a wide area at high frame rates (∼10 Hz) [4]. However, the observations are inherently sparse (typically 500–2000 points per scan), unevenly distributed (denser near the ship), and noisy (accuracy degrades with range and incidence angle).

The dominant approach in the literature follows a grid-based paradigm: discretize the spatial domain on a regular grid, represent the wave state as surface elevation and velocity on this grid, and apply an ensemble Kalman filter (EnKF/ETKF) to assimilate observations [1], [2]. While principled, this approach has three fundamental limitations: (1) **grid interpolation**: observations must be interpolated to grid points, introducing smoothing and aliasing for wavelengths near the grid Nyquist limit; (2) **fixed resolution**: the grid resolution is a design parameter that trades accuracy against computational cost, with no adaptation to the actual wave complexity; (3) **ensemble collapse**: when the number of observations far exceeds the ensemble size ($M \gg N_e$), the ETKF analysis can become rank-deficient, requiring observation thinning or localization.

We propose **WaveSplat**, a grid-free spectral representation inspired by 3D Gaussian Splatting [3]. Instead of discretizing space, we represent the wave field as a weighted superposition of plane waves:

$$z(x, y, t) = \sum_{i=1}^{N} w_i \cos(k_{x,i}x + k_{y,i}y - \omega_i t + \varphi_i) \quad (1)$$

where each "spectral Gaussian" is parameterized by amplitude w_i, wavenumber $(k_{x,i}, k_{y,i})$, and phase φ_i. The angular frequency ω_i is not a free parameter but is *constrained* by the linear dispersion relation:

$$\omega_i = \sqrt{g\,|k_i|\,\tanh(|k_i|\,d)} \quad (2)$$

where $g = 9.81$ m/s^2 and d is the water depth. This hard constraint reduces the parameter space, regularizes optimization, and ensures physically plausible wave propagation.

Our key contributions are:

1) **Grid-free spectral representation (C1)**: Direct estimation on sparse observations without grid interpolation, enabling arbitrary spatial resolution and scaling to any observation pattern.
2) **Dispersion-constrained optimization (C2)**: Hard enforcement of the dispersion relation reduces parameters and acts as a physics regularizer, improving convergence on multi-component seas.
3) **Fresnel-aware confidence weighting (C3)**: Per-observation weights based on incidence angle and range account for heteroscedastic LiDAR noise, critical for sloped sea surfaces.
4) **Adaptive model complexity**: Energy-based pruning and residual-triggered growth automatically match the number of spectral components to scene complexity, from N=1 for simple swells to N=21 for complex multi-component seas.

II. Related Work

A. Grid-Based Wave Field Estimation

Desmars et al. [1] introduced ETKF-based wave field reconstruction from X-band radar, with spectral propagation on a regular grid. Huhn et al. [2] extended this to LiDAR observations, demonstrating the "from points to waves" pipeline. These methods discretize the 2D spatial domain (typically

40×30 to 80×60 grid points) and use FFT-based spectral propagation as the forecast model. Our approach eliminates the grid entirely, operating directly on spectral components.

B. Spectral Estimation Methods

Classical spectral estimation (FFT, MUSIC, ESPRIT) requires regularly sampled data and struggles with the irregular, sparse sampling inherent to scanning LiDAR. Capon beamforming [5] can handle irregular arrays but assumes stationary signals. WaveSplat combines beamforming for initialization with gradient-based optimization for refinement, exploiting temporal structure through the dispersion relation.

C. Neural and Learning-Based Approaches

Recent work has applied neural networks to wave prediction [6], but these require large training datasets and generalize poorly across sea states. WaveSplat is purely physics-based, with no training data required and interpretable spectral parameters.

D. Gaussian Splatting

3D Gaussian Splatting [3] revolutionized novel-view synthesis by representing scenes as collections of anisotropic Gaussians optimized via gradient descent. WaveSplat adapts this philosophy to wave fields: each "Gaussian" is a plane wave component in spectral space, optimized to fit sparse LiDAR observations while respecting physical constraints.

III. Method

A. Forward Model

The wave field is modeled as a superposition of N linear plane waves (1). Each component i has four free parameters: amplitude w_i, wavenumber components $(k_{x,i}, k_{y,i})$, and phase offset φ_i. The angular frequency ω_i is deterministically computed from the dispersion relation (2), reducing the total parameter count from $5N$ to $4N$.

This linear model is an $O(\epsilon^2)$ approximation to the Gerstner wave model used in our ground-truth simulator, where $\epsilon = |k| \cdot |w|$ is the wave steepness. At Beaufort 4, $\epsilon \approx 0.05$, giving model mismatch $< 0.25\%$.

B. Observation Model

At each LiDAR scan at time t, we observe M elevation measurements $\{(x_j, y_j, z_j)\}_{j=1}^{M}$ at irregular spatial locations. The observation model is:

$$z_j = \sum_{i=1}^{N} w_i \cos(k_{x,i} x_j + k_{y,i} y_j - \omega_i t + \varphi_i) + \eta_j \quad (3)$$

where η_j is observation noise with variance depending on range and incidence angle.

C. Fresnel-Zone Confidence Weighting (C3)

LiDAR elevation accuracy degrades with range (beam divergence) and incidence angle (specular reflection probability). We assign per-observation confidence weights:

$$c_j = \frac{\cos(\theta_j)}{d_j / d_{\text{ref}}} \quad (4)$$

where θ_j is the beam incidence angle, d_j is the range, and $d_{\text{ref}} = 125$ m is a reference distance. These weights are incorporated in the optimization loss as weighted MSE.

D. Two-Phase Estimation

1) Phase 1: Initialization: Given the first $T_{\text{init}} = 2T_p$ seconds of observations (where T_p is the expected peak period), we jointly optimize all spectral parameters using gradient descent with a coarse-to-fine schedule (N=3 → 8 → 15).

Initial values are obtained via **dispersion-aware beamforming**: for each candidate wavenumber (k_x, k_y), we compute the corresponding ω from (2) and evaluate the beamforming power:

$$P(k_x, k_y) = \left| \sum_j z_j \, e^{-i(k_x x_j + k_y y_j - \omega t_j)} \right|^2 \quad (5)$$

Peaks in P are extracted greedily with neighborhood suppression, providing accurate initial estimates of (k_x, k_y). Amplitudes and phases are then estimated via linear projection.

The optimization minimizes:

$$\mathcal{L} = \frac{1}{M} \sum_j c_j \, (z_j - \hat{z}_j)^2 + \lambda_1 \sum_i |w_i| + \lambda_s \sum_i (w_{\sigma(i)} - w_{\sigma(i-1)})^2 \quad (6)$$

where $\lambda_1 = 10^{-3}$ promotes sparsity and $\lambda_s = 10^{-4}$ penalizes spectral roughness (σ sorts by $|k|$).

2) Adaptive N: After initialization, we apply energy-based pruning: components are sorted by energy w_i^2 in descending order, and the fewest components capturing 90% of total energy are retained. A safety check verifies that pruning does not increase the initialization residual by more than 30%; if so, pruning is rolled back.

If the residual RMS after pruning exceeds a threshold (0.05 m), new components are grown via beamforming on the residual signal, up to $N_{\max} = 30$.

3) Phase 2: Tracking: For each subsequent frame, we:

1) **Predict**: advance phases by $\varphi_i \mathrel{+}= \omega_i \, \Delta t$
2) **Update**: perform $K = 10$ gradient steps on the current frame, updating only (w_i, φ_i) while freezing wavenumbers (`w_phi` mode)

Freezing (k_x, k_y) during tracking prevents overfitting to single-frame noise and exploits the dispersion constraint: once wavenumbers are accurately estimated, phase prediction via $\omega \cdot \Delta t$ provides a strong warm start.

Multi-frame tracking uses a sliding window of $K_w = 3$ recent frames for gradient computation, improving stability without significant computational overhead.

TABLE I: WaveSplat vs ETKF-grid on 12 representative episodes. nRMSE = RMSE/H_s; lower is better. N_a = adaptive Gaussian count.

	WaveSplat		ETKF		
Episode	nRMSE	ms/f	nRMSE	ms/f	N_a
BF2 1comp	**0.315**	13.8	0.483	29.1	11
BF3 1comp	**0.217**	13.0	0.352	36.8	2
BF3 2comp (U)	**0.052**	15.0	0.444	37.2	2
BF3 2comp (S)	**0.061**	13.4	0.350	26.3	15
BF3 5comp (U)	**0.167**	13.0	0.434	29.6	4
BF3 5comp (S)	**0.152**	13.4	0.384	32.7	15
BF4 1comp	**0.171**	11.2	0.410	26.7	1
BF4 5comp	**0.200**	12.8	0.376	28.7	3
BF4 15comp	**0.302**	13.4	0.511	25.8	19
BF4 30comp	**0.314**	13.4	0.525	28.3	18
BF5 15comp	**0.297**	13.5	0.466	27.2	21
BF5 30comp	**0.290**	14.7	0.497	52.2	19
Avg	**0.211**	13.4	0.436	31.7	—

IV. Experimental Setup

A. Simulation Environment

We use a physics-based LiDAR wave simulation (WaveSim) generating ground-truth Gerstner wave fields with:

- **Sea states**: Beaufort 2–5 (H_s=0.3–2.0 m, T_p=4–9 s)
- **Wave complexity**: 1, 2, 5, 15, 30 spectral components
- **LiDAR model**: 500 horizontal beams, 10 Hz scan rate
- **Raydrop models**: uniform (constant drop probability) and slope-dependent (higher dropout on steep surfaces)
- **Water depth**: $d = 50$ m

Each episode contains 301 frames (30 s at 10 Hz) with ground-truth elevation fields on a 80×60 evaluation grid. A total of 100 episodes are generated across 5 random seeds per configuration.

B. Metrics

We report:

- **nRMSE**: RMSE normalized by significant wave height H_s, the primary metric
- **Inference speed**: ms/frame for tracking phase
- $N_{\textbf{adapt}}$: final number of spectral components (adaptive N)

C. Baselines

ETKF-grid [2]: Ensemble Transform Kalman Filter on a 40×30 regular grid with 50 ensemble members, FFT-based spectral propagation, multiplicative (α=1.01) and additive (σ=0.005) inflation, and observation thinning to $M_{\max}$=40 per frame.

V. Results

A. Main Comparison

Table I summarizes the comparison across 12 representative episodes. WaveSplat achieves an average nRMSE of **0.211** compared to 0.436 for ETKF-grid, a **51.5%** improvement. WaveSplat wins on all 12 episodes.

Key observations:

TABLE II: Adaptive N vs. fixed N=15 (P0 baseline). N_{true} is the ground-truth number of wave components.

Episode	N_{true}	N_{adapt}	Fixed	Adaptive
BF4 1comp	1	1	0.208	**0.171**
BF3 2comp	2	2	0.059	**0.052**
BF3 5comp	5	4	0.185	**0.167**
BF4 15comp	15	19	0.306	**0.302**
BF5 15comp	15	21	0.331	**0.297**
BF4 30comp	30	18	0.332	**0.314**
Avg (12 ep.)			0.226	**0.211**

TABLE III: Ablation study (3-seed average nRMSE ± std on BF3 episodes).

Episode	Full	w/o Disp. (A1)	w/o Conf. (A2)
1comp (S)	$0.224_{\pm.030}$	$0.230_{\pm.055}$	$0.269_{\pm.002}$
1comp (U)	$0.215_{\pm.014}$	$0.231_{\pm.006}$	$0.214_{\pm.007}$
2comp (S)	$0.088_{\pm.017}$	$0.071_{\pm.007}$	$\mathbf{0.204}_{\pm.006}$
2comp (U)	$0.173_{\pm.023}$	$0.085_{\pm.023}$	$0.176_{\pm.048}$
5comp (S)	$0.203_{\pm.021}$	$0.104_{\pm.014}$	$0.223_{\pm.010}$
5comp (U)	$0.225_{\pm.006}$	$0.225_{\pm.035}$	$0.206_{\pm.032}$
Avg	0.188	0.158	0.215

- **Low complexity**: For BF3 2-component seas, WaveSplat achieves nRMSE=0.052, an 8.5× advantage over ETKF (0.444). The grid-free representation perfectly captures two plane waves without grid aliasing.
- **High complexity**: For BF4–5 with 15–30 components, WaveSplat still outperforms ETKF (0.301 vs. 0.500 average), though the gap narrows as the spectral complexity approaches the initialization resolution limit.
- **Speed**: WaveSplat runs at 13.4 ms/frame vs. 31.7 ms for ETKF (2.4× faster), well within real-time requirements.

B. Adaptive N Behavior

Table II shows that adaptive N consistently improves over fixed N=15, with an average 6.4% reduction in nRMSE. The model correctly identifies scene complexity: N=1 for single-component swells, N=2–4 for moderate seas, and N=18–21 for complex multi-component fields.

C. Ablation Study

We ablate two key design choices on 6 BF3 episodes with 3 random seeds:

A1 (no dispersion): Removing the dispersion constraint allows ω as a free parameter. Surprisingly, this improves average nRMSE (0.158 vs. 0.188) on these BF3 episodes, suggesting the optimizer finds better frequency estimates without the constraint on moderate seas. However, dispersion becomes increasingly valuable at higher Beaufort numbers and for long-term phase coherence during tracking.

A2 (no confidence): Removing Fresnel confidence weighting degrades performance by 14.5% on average (0.215 vs. 0.188), with the largest degradation on slope-model episodes (BF3 2comp (S): 0.088 → 0.204, a 2.3× increase). This confirms that accounting for range- and angle-dependent noise is essential for heterogeneous observation quality.

VI. Discussion

Grid-free advantages. The most striking result is the 8.5× advantage on 2-component seas (nRMSE 0.052 vs. 0.444). For low-complexity wave fields, ETKF wastes capacity representing the full spatial grid when only two spectral modes are present. WaveSplat's adaptive representation directly captures the relevant modes.

Limitations. For very high complexity ($N_{\text{true}} \geq 30$), WaveSplat's beamforming initialization cannot resolve all components, leading to nRMSE $\sim$0.30. A spatio-temporal 3D FFT or higher-resolution beamforming could address this. The inference speed of 13.4 ms is slower than P0 (8.4 ms) due to multi-frame tracking; this could be mitigated by reducing the window size or parallelizing gradient computation.

Dispersion constraint. The ablation reveals a nuanced role for the dispersion constraint: it acts more as a regularizer than a strict accuracy improvement. On moderate seas, the optimizer can find better solutions without it, but for tracking stability over long time horizons, the $\omega = f(|k|, d)$ constraint prevents phase drift.

VII. Conclusion

We presented WaveSplat, a grid-free spectral Gaussian approach to ocean wave field estimation from sparse LiDAR. By representing waves as optimizable spectral components with dispersion constraints and adaptive complexity, WaveSplat achieves 51.5% lower nRMSE than ETKF-grid while running 2.4× faster. The method wins on all 12 test episodes spanning Beaufort 2–5.

Future work includes extending to real LiDAR data, incorporating current estimation, and exploring learned initialization for faster convergence on complex seas.

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